What Justifies Splitting dy/dx?

On the Meaning of Differentials in Exact Equations

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1 The Puzzle

Consider the chain-rule identity for $F(x, y)$ along a curve $y = y(x)$:

$$\frac{dF}{dx} = \left.\frac{\partial F}{\partial x}\right|_{(x,y(x))} + \left.\frac{\partial F}{\partial y}\right|_{(x,y(x))} \frac{dy}{dx}.$$

This equation underlies the theory of exact differential equations $M\,dx + N\,dy = 0$. Yet its use raises an old discomfort. Elementary calculus insists, correctly, that dy/dx is *not* a fraction but a single symbol denoting a limit. Later, when solving equations of the form $N\,dy/dx = -M$, we are casually told to "multiply through by dx" and write $M\,dx + N\,dy = 0$, treating dy/dx exactly like a fraction after all. What justifies this? And underneath the notation, what actually *is* a differential?

2 The Chain Rule Is Not Fraction-Splitting

It is worth first noticing that the identity above does not, by itself, split anything. The multivariable chain rule

$$\frac{dF}{dx} = \left.\frac{\partial F}{\partial x}\right|_{(x,y(x))} + \left.\frac{\partial F}{\partial y}\right|_{(x,y(x))} \frac{dy}{dx}$$

treats dy/dx as one indivisible object: the ordinary derivative of y with respect to x, valid because y is assumed to be a differentiable function of x along the curve in question. No fraction is decomposed here; this is simply differentiation of a composite function.

The real manipulation occurs one step later. Starting from

$$\frac{dF}{dx} = M + N\frac{dy}{dx} = 0,$$

we rearrange to $dy/dx = -M/N$, and then "cross-multiply" to obtain

$$M\,dx + N\,dy = 0.$$

This last move is the one that needs justifying.

3 Two Objects, One Notation

The resolution is that calculus silently uses the symbol dy/dx for two different mathematical objects, and the discomfort comes from not distinguishing them.

The derivative

This is

$$\frac{dy}{dx} := \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}, \qquad \Delta y := f(x + \Delta x) - f(x),$$

a single limiting value. As a limit, it is genuinely not a fraction: both Δy and Δx vanish in the limiting process, and writing the result as "dy/dx" is a historical notation, not a literal quotient of two numbers called dy and dx. This is precisely what elementary courses warn against, and the warning is correct.

The differential

Once the derivative $f'(x)$ exists, one may *define* two new quantities. Let dx be an arbitrary nonzero real number—an independent variable, not required to be small—and define

$$dy := f'(x)\, dx.$$

Now dy and dx are genuine real numbers related by an ordinary algebraic equation. Dividing both sides by dx gives $dy/dx = f'(x)$, trivially, by cancellation, precisely because dy was constructed that way. This dy/dx *is* a fraction, and cross-multiplying it is ordinary algebra.

The two objects are built to agree numerically, which is why they share notation. But logically, the differential is manufactured *after* the derivative exists, specifically so that their ratio reproduces it. The warning "it's not a fraction" applies to the derivative. The cross-multiplication trick used in solving $M\, dx + N\, dy = 0$ is legitimate because at that point the discussion has quietly shifted to the differential, without the shift being announced.

What is still missing is a precise statement of *what kind of object* each side of this shift actually is—not just the equation it satisfies, but its type: its domain, its codomain, and the properties that come with it. Section 5 makes this explicit.

4 What a Differential Actually Is

There are three standard rigorous accounts of the differential, of increasing sophistication.

(a) Linear approximation

Here $dy = f'(x)\, dx$ is defined as the *linear part* of the true increment $\Delta y = f(x + dx) - f(x)$. Precisely,

$$\Delta y = dy + \varepsilon(dx) \cdot dx, \qquad \varepsilon(dx) \to 0 \text{ as } dx \to 0.$$

So dy is not the actual change in y, but the tangent line's prediction of that change. This is the sense in which dx and dy are genuine numbers once a point and an increment are fixed.

(b) Linear map

The modern, coordinate-free definition treats the differential of $F(x, y)$ at a point as the linear map (see Appendix A if the terms *linear map* and *tangent space* below are unfamiliar)

$$dF_{(x,y)}(h, k) = \frac{\partial F}{\partial x}(x, y)\, h + \frac{\partial F}{\partial y}(x, y)\, k,$$

where dx and dy are themselves defined as the coordinate projections $dx(h,k) = h$ and $dy(h,k) = k$—literally the differentials of the coordinate functions, which is why the names are consistent rather than coincidental. Then

$$dF = \frac{\partial F}{\partial x}\,dx + \frac{\partial F}{\partial y}\,dy$$

is an exact identity between linear functionals: no infinitesimals, no hand-waving, only linear algebra. It expresses dF in the dual basis $\{dx, dy\}$. Evaluating both sides on the tangent vector $(h, y'(x)h)$ of a curve $y = y(x)$ and dividing by h recovers the chain-rule identity directly—not by splitting a fraction, but by evaluating a linear map on a vector.

(c) Infinitesimal

Non-standard analysis takes dx to be an actual nonzero infinitesimal hyperreal number, and defines $dy := f(x + dx) - f(x)$ as the true increment, computed within the hyperreals. Then dy/dx is a literal quotient of hyperreal numbers, infinitely close to $f'(x)$:

$$\operatorname{st}\left(\frac{dy}{dx}\right) = f'(x).$$

This is arguably the most faithful formalization of how Leibniz and Euler actually reasoned: dx really is a number, just not a standard real one.

5 The Type Signatures of Differential Objects

The three accounts above all write down a symbol dx and a symbol dy, and all three relate them by some version of $dy = f'(x)\,dx$. But dx in framework (a) and dx in framework (b) are not the same kind of mathematical object merely dressed differently—they have different domains and codomains outright. Stating each object's type, in the sense of a function signature (inputs in, output out), is what finally pins down what a differential *is* in a given framework, as opposed to what equation it satisfies.

Objects common to all three frameworks

Before the frameworks diverge, some notation means the same thing throughout.

- x: a real number, a point of the domain where f is being differentiated. (In §6 below, x is reused for the first coordinate function $x : \mathbb{R}^2 \to \mathbb{R}$; context disambiguates the two uses.)
- f: a function $f : I \to \mathbb{R}$, where $I \subseteq \mathbb{R}$ is an open interval, differentiable at the point x in question.
- $f'(x)$: for fixed x, a real number—the value of the derivative at that point. Letting x vary, $f' : I \to \mathbb{R}$ is itself a function, related to f by the usual linearity, product, and chain rules.

Everything that follows is about what *additional* object, named dx or dy, each framework builds on top of this common ground.

(a) Linear approximation: dx and dy as numbers

Signature 5.1. $dx \in \mathbb{R}$; $\quad dy = f'(x)\,dx \in \mathbb{R}$.

Here dx is not a function of x at all: it is an independent real number, freely chosen, playing the role of an input increment. Given x and dx, the differential dy is the value of the map

$$\mathrm{d}f : I \times \mathbb{R} \to \mathbb{R}, \qquad (x, dx) \mapsto f'(x)\, dx,$$

which is linear in its second argument for each fixed x. Ordinary notation shows only "dx" on the page and lets the dependence on x stay implicit in "the point where we're differentiating"—but dy genuinely depends on *two* numbers, not one.

Symbol	**Type**	**Domain / Codomain**	**Key property**
x	point	$x \in I$	fixed base point
dx	number	$dx \in \mathbb{R}$, arbitrary, $\neq 0$	independent variable
dy	number	$dy \in \mathbb{R}$	$dy = f'(x)\, dx$; linear in dx

(b) Linear map: dx and dy as functionals on tangent vectors

Signature 5.2. $dx_p, dy_p : T_p\mathbb{R}^2 \to \mathbb{R}; \quad dF_p = \dfrac{\partial F}{\partial x}(p)\, dx_p + \dfrac{\partial F}{\partial y}(p)\, dy_p : T_p\mathbb{R}^2 \to \mathbb{R}.$

Fix a point $p = (x, y) \in \mathbb{R}^2$ and its tangent space $T_p\mathbb{R}^2 \cong \mathbb{R}^2$, whose elements are vectors (h, k). Here dx_p and dy_p are not numbers but *linear functionals*—elements of the cotangent space $T_p^*\mathbb{R}^2$—namely the coordinate projections $dx_p(h, k) = h$ and $dy_p(h, k) = k$. The partial derivatives $\partial F/\partial x, \partial F/\partial y : \mathbb{R}^2 \to \mathbb{R}$ are ordinary scalar-valued functions of the point p (they eat a point, return a number), whereas dF_p is again a linear functional on $T_p\mathbb{R}^2$, built as the linear combination above. Letting p vary, dF is a *section of the cotangent bundle* $T^*\mathbb{R}^2$—a 1-form, i.e. an assignment of a functional to every point, not a single functional or a single number. The same goes for $M, N : \mathbb{R}^2 \to \mathbb{R}$ (scalar functions of p) versus $\omega = M\, dx + N\, dy$ (a 1-form, $\omega_p = M(p)\, dx_p + N(p)\, dy_p$).

Symbol	**Type**	**Domain / Codomain**	**Key property**
$p = (x, y)$	point	$p \in \mathbb{R}^2$	evaluation point
$\partial F/\partial x$	scalar field	$\mathbb{R}^2 \to \mathbb{R}$	function of p alone
dx_p, dy_p	linear functional	$T_p\mathbb{R}^2 \to \mathbb{R}$	coordinate projections
dF_p	linear functional	$T_p\mathbb{R}^2 \to \mathbb{R}$	$\frac{\partial F}{\partial x}(p)\, dx_p + \frac{\partial F}{\partial y}(p)\, dy_p$
dF, ω	1-form (section)	$p \mapsto (T_p\mathbb{R}^2 \to \mathbb{R})$	functional at *every* point

(c) Infinitesimal: dx and dy as hyperreal numbers

Signature 5.3. $dx \in \mathbb{R}^*$ infinitesimal; $\quad dy = {}^*f(x + dx) - {}^*f(x) \in \mathbb{R}^*; \quad \mathrm{st}(dy/dx) = f'(x) \in \mathbb{R}.$

Let $\mathbb{R}^*$ be the hyperreal field extending $\mathbb{R}$. Here $dx \in \mathbb{R}^*$ is a genuine, fixed, nonzero number—just an infinitesimal one, meaning $|dx| < r$ for every positive real r. The function f extends, via the transfer principle, to ${}^*f : \mathbb{R}^* \to \mathbb{R}^*$, and dy is *defined* as the true increment $dy := {}^*f(x + dx) - {}^*f(x) \in \mathbb{R}^*$, itself infinitesimal when f is continuous at x. The quotient dy/dx is an ordinary quotient of hyperreal numbers, living in $\mathbb{R}^*$; generically it is *finite* (not infinitesimal), which is what allows the standard-part map st, defined on finite hyperreals, $\mathrm{st} : \mathrm{fin}(\mathbb{R}^*) \to \mathbb{R}$, to send it back to an ordinary real number equal to $f'(x)$.

Symbol	Type	Domain / Codomain	Key property
dx	hyperreal number	$dx \in \mathbb{R}^*$, infinitesimal $\neq 0$	$\lvert dx\rvert < r$, all real $r > 0$
${}^*f$	function	$\mathbb{R}^* \to \mathbb{R}^*$	transfer of f
dy	hyperreal number	$dy \in \mathbb{R}^*$	$:= {}^*f(x + dx) - {}^*f(x)$
dy/dx	hyperreal number	$\in \text{fin}(\mathbb{R}^*) \subset \mathbb{R}^*$	finite, not infinitesimal
st	function	$\text{fin}(\mathbb{R}^*) \to \mathbb{R}$	$\text{st}(dy/dx) = f'(x)$

Same notation, different type

The punchline of this section is visible only by comparing rows: three frameworks reuse the letters dx and dy for objects that do not even live in the same kind of set.

Framework	Type of dx	Type of dy
(a) Linear approximation	number, $dx \in \mathbb{R}$	number, $dy \in \mathbb{R}$
(b) Linear map	functional, $dx_p \in T_p^*\mathbb{R}^2$	functional, $dy_p \in T_p^*\mathbb{R}^2$
(c) Infinitesimal	hyperreal number, $dx \in \mathbb{R}^*$	hyperreal number, $dy \in \mathbb{R}^*$

Only within framework (c) is dx literally a nonzero number that dy is divided by to form a bona fide quotient; in (a) the "fraction" dy/dx is trivial cancellation by construction; in (b) there is no division at all, since dx_p and dy_p are functionals, not numbers, and dF/dx is recovered by *evaluating* dF_p on a tangent vector, not by dividing.

6 Why Exact Equations Are Best Written as $M\,dx + N\,dy = 0$

Framing (b) reveals the deeper reason for this notation, beyond mere convenience. Regard

$$\omega = M\,dx + N\,dy$$

as a *differential* 1*-form*: at each point (x, y) it is the linear functional $\omega_{(x,y)}(h, k) = M(x, y)\,h + N(x, y)\,k$ acting on tangent vectors. The equation $\omega = 0$ asserts that along a solution curve with velocity $(x'(t), y'(t))$,

$$M\,x'(t) + N\,y'(t) = 0.$$

Parametrizing the curve by x itself, so that $t = x$, this becomes $M + N\,dy/dx = 0$—the original ODE. The "cross-multiplication" is thus not a trick applied to a fraction; it is the coordinate-free statement $\omega = 0$, read off in the x-parametrization. Writing $M\,dx + N\,dy = 0$ instead of $dy/dx = -M/N$ is deliberately symmetric in x and y: it does not presuppose which variable is a function of the other, which matters when one later solves for $x(y)$ instead of $y(x)$.

This symmetry is also why the exactness criterion,

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x},$$

works as it does: it is precisely the statement that ω is a *closed* form. By the Poincaré lemma, a closed 1-form on a simply connected domain is *exact*, meaning $\omega = dF$ for some potential F, in exactly the sense of (b) above. The name "exact differential equation" is therefore not a metaphor—it invokes the exact/closed distinction for differential forms directly.

7 Conclusion

Elementary calculus is right that the derivative, defined as a limit, is not a fraction. But the differential is a distinct, later-introduced notion: a real number (or a linear functional, or an infinitesimal, depending on the chosen framework) satisfying $dy = f'(x)\,dx$ by construction rather than by limiting process. Once this relationship is taken as a definition, dividing or cross-multiplying it is ordinary algebra. The apparent contradiction dissolves once one recognizes that calculus reuses a single piece of notation, dy/dx, for two related but logically distinct objects, and that the switch between them is rarely announced.

What Section 5 adds is that this is not merely a two-way ambiguity between "derivative" and "differential." Even within "the differential," the type of dx and dy—number, linear functional, or hyperreal number—depends on which of the three frameworks is in force. Writing down the equation $dy = f'(x)\,dx$ is not enough to say what dx *is*; only its domain and codomain, stated explicitly, do that.

A Linear Maps and Tangent Spaces

Framework (b) in Section 5 leans on two ideas that elementary calculus does not usually name explicitly: *linear maps* and *tangent spaces*. Both are simpler than their names suggest.

Linear maps

Definition A.1. A function $L : \mathbb{R}^2 \to \mathbb{R}$ is a *linear map* (or *linear functional*) if, for all $(h_1, k_1), (h_2, k_2) \in \mathbb{R}^2$ and all scalars $c \in \mathbb{R}$,

$$L(h_1 + h_2,\, k_1 + k_2) = L(h_1, k_1) + L(h_2, k_2), \qquad L(ch_1, ck_1) = c\,L(h_1, k_1).$$

In down-to-earth terms, a linear map out of $\mathbb{R}^2$ is nothing more exotic than an expression

$$L(h, k) = a\,h + b\,k$$

for fixed constants $a, b \in \mathbb{R}$—and every such expression is automatically linear, so this formula is not a special case but the *general* one. Contrast it with an affine function like $L(h, k) = ah+bk+c$ with $c \neq 0$: adding a constant term destroys linearity, since a linear map must send $(0, 0) \mapsto 0$, while $L(0, 0) = c \neq 0$ here.

Example A.2. $L(h, k) = 3h - 2k$ is linear: $L(1, 0) = 3$, $L(0, 1) = -2$, and $L(2, 5) = 6 - 10 = -4$, which agrees with $2L(1, 0) + 5L(0, 1) = 6 - 10 = -4$, as linearity requires.

The two simplest linear maps on $\mathbb{R}^2$ just read off a coordinate: $\pi_1(h, k) = h$ and $\pi_2(h, k) = k$. These are exactly dx_p and dy_p from Section 5(b): as *formulas* they do not even mention p, which is why the subscript is there only to record *which* tangent space they are regarded as acting on (see below). Every other linear map on $\mathbb{R}^2$, such as $dF_p(h, k) = F_x(p)\,h + F_y(p)\,k$, is a constant-coefficient combination $a\pi_1 + b\pi_2$ of these two—this is what Section 5(b) means by expressing dF "in the dual basis $\{dx, dy\}$."

Tangent vectors and the tangent space

Fix a point $p = (x_0, y_0) \in \mathbb{R}^2$ and a curve through p: differentiable functions $x(t), y(t)$ with $x(t_0) = x_0$, $y(t_0) = y_0$ at some parameter value t_0. Its *velocity vector* at p is

$$(x'(t_0),\, y'(t_0)) \in \mathbb{R}^2,$$

the instantaneous direction and speed of motion through p. Many curves through p can share one velocity vector, and every pair $(h,k) \in \mathbb{R}^2$ is attained by some curve through p—e.g. the straight line $x(t) = x_0 + th$, $y(t) = y_0 + tk$ at $t_0 = 0$. So the set of velocity vectors attainable at p is all of $\mathbb{R}^2$.

Definition A.3. The *tangent space* to $\mathbb{R}^2$ at p, written $T_p\mathbb{R}^2$, is the set of velocity vectors of curves through p. As a set it is identical to $\mathbb{R}^2$, but its elements are pictured as *arrows based at* p—directions and speeds of motion—rather than as points or positions.

Why distinguish $T_p\mathbb{R}^2$ from $\mathbb{R}^2$ if they are the same set? Because "a location" and "a direction of travel through that location" are conceptually different, even though both happen to be pairs of numbers in the flat plane. The distinction is invisible in $\mathbb{R}^2$ precisely because the plane is flat: on a curved surface (a sphere, say) the tangent space at a point is a flat plane genuinely different from the surface itself, and tangent spaces at two different points cannot be identified with each other. Framework (b) writes $T_p\mathbb{R}^2$, not just $\mathbb{R}^2$, so that its notation carries over unchanged to that curved setting—though for this essay $T_p\mathbb{R}^2$ can always be read simply as "$\mathbb{R}^2$, but thought of as velocity vectors at p."

A linear functional on the tangent space—an element of $T_p^*\mathbb{R}^2$—is a linear map that eats a tangent vector $(h,k) \in T_p\mathbb{R}^2$ and returns a number. This is precisely the type of dx_p, dy_p, and dF_p in Section 5(b): each turns a direction of travel through p into a single real number.

Example A.4. Let $p = (1,1)$ and take the curve $x(t) = t$, $y(t) = t^2$, through p at $t = 1$ with velocity $(x'(1), y'(1)) = (1,2) \in T_p\mathbb{R}^2$. For $F(x,y) = x^2y$, $F_x(p) = 2xy|_p = 2$ and $F_y(p) = x^2|_p = 1$, so

$$dF_p(1,2) = F_x(p) \cdot 1 + F_y(p) \cdot 2 = 2 \cdot 1 + 1 \cdot 2 = 4.$$

This matches the ordinary chain rule directly: $\frac{d}{dt}F(x(t), y(t))\big|_{t=1} = \frac{d}{dt}(t^4)\big|_{t=1} = 4t^3\big|_{t=1} = 4$. Evaluating the linear functional dF_p on the tangent vector $(1,2)$ *is* evaluating this derivative—the general fact Section 5(b) uses to recover the chain rule without splitting a fraction.

Self-Check Questions

1. In the identity $\dfrac{dF}{dx} = \dfrac{\partial F}{\partial x} + \dfrac{\partial F}{\partial y}\dfrac{dy}{dx}$, is dy/dx being split into two fractions? Justify your answer.
2. State the precise definition of dy/dx as a derivative (the limit definition), and explain why this object cannot literally be a quotient of two numbers.
3. State the definition of the differential dy in terms of dx, once $f'(x)$ is known. Why does dividing this equation by dx trivially recover $f'(x)$?
4. Explain, in your own words, why the derivative and the differential can share the same notation dy/dx without being the same kind of object.
5. Under the linear-approximation view of the differential, is dy equal to Δy? What is the precise relationship between them?
6. In the linear-map definition, what are dx and dy, formally? What role do they play as elements of a dual basis?
7. Using the linear-map definition, derive the chain-rule identity $dF/dx = \partial F/\partial x + (\partial F/\partial y)(dy/dx)$ without invoking fractions at any point.

8. What is an infinitesimal hyperreal, and how does the non-standard analysis definition of dy/dx differ from the limit definition? What operation recovers $f'(x)$ from the hyperreal quotient?

9. Why is $M\,dx+N\,dy = 0$ considered a more natural way to state an ODE than $dy/dx = -M/N$? What symmetry does it preserve?

10. Explain the geometric meaning of a differential 1-form $\omega = M\,dx + N\,dy$, and what it means for ω to vanish along a curve.

11. State the exactness condition $\partial M/\partial y = \partial N/\partial x$ and explain, using the Poincaré lemma, why it guarantees the existence of a potential function F with $dF = \omega$.

12. Why is the term "exact" in "exact differential equation" not merely descriptive but a direct reference to the exact/closed distinction for differential forms?

13. Suppose someone tells you "dy/dx is just a fraction, you can always cross-multiply." Identify precisely where this claim is true and where it requires qualification.

14. In framework (a), is dy a function of one variable or two? Name them, and explain why ordinary notation only displays one.

15. In framework (b), is dx_p a number or a function? State its domain and its codomain precisely.

16. Still in framework (b), what is the difference in type between $\partial F/\partial x$ and dF_p? What is the difference in type between dF_p (fixed p) and dF (as p varies)?

17. In framework (c), what is the codomain of the standard-part map st, and why must dy/dx be *finite* (as opposed to infinite or infinitesimal) for $\mathrm{st}(dy/dx)$ to be defined?

18. Compare the three frameworks' treatment of dx: in which one, if any, is dx literally a nonzero number that dy can be divided by to form a genuine quotient? What replaces division in the other framework(s)?

19. Explain why stating the equation $dy = f'(x)\,dx$ alone does not pin down what kind of object dx is, and why the domain/codomain of dx must be specified separately.

20. Give the general formula for a linear map $L : \mathbb{R}^2 \to \mathbb{R}$, and explain why an affine function $L(h, k) = ah + bk + c$ with $c \neq 0$ fails to be linear.

21. What is a tangent vector at a point $p \in \mathbb{R}^2$, and why is the tangent space $T_p\mathbb{R}^2$ conceptually different from $\mathbb{R}^2$ even though the two sets coincide?